

Modeling the Impact of Agrivoltaics on Corn Yield and Water Balance in the US

Uwe R. Kortshagen^{1,*}  and Vivian E. Ferry² 

¹Department of Mechanical Engineering, University of Minnesota, Minneapolis, MN, USA

²Department of Chemical Engineering and Materials Science, University of Minnesota, Minneapolis, MN, USA

*Correspondence: Uwe R. Kortshagen, korts001@umn.edu

Abstract. Corn farming occupies a substantial portion of agricultural land in the United States, yet the potential impacts of agrivoltaic systems on commodity crop yields remain poorly quantified at a national scale. Building upon our previous county-level modeling framework, we extended our approach to analyze the top 1000 corn-producing counties in the U.S. For each county, the model was calibrated by comparing 20 years of USDA-reported yield data with model predictions to reflect local growing conditions, management practices, and genetic improvements. Our analysis quantified the trade-offs between crop yield loss due to shading from sparse solar arrays and potential yield gains resulting from increased soil moisture retention under partial shading. The results indicate that corn yield decreases by approximately 0.55% per 1% shading, while drought stress, as measured by the Agricultural Reference Index for Drought (ARID), decreases by approximately 0.8% per 1% shading due to reduced evapotranspiration. The model also accurately captured the historical ~0.5% per year yield increase attributed to genetic improvements and advanced management practices. Furthermore, our findings suggest that modern corn varieties exhibit substantially higher drought resistance compared to traditional cultivars, such as McCurdy 84aa, used in previous crop models.

These results provide a comprehensive, county-level assessment of the viability of agrivoltaic systems in U.S. corn production, highlighting the need to balance energy generation with minimal crop yield impacts. The framework established here can inform future agrivoltaic design, policy planning, and economic evaluations for large-scale dual-use deployment on commodity crop farmland.

Keywords: Agrivoltaics, Crop Modeling, Commodity Crops, Drought Stress, County-Level Analysis, Solar Shading Effects

1. Introduction

The dual use of agricultural land for both crop production and photovoltaic (PV) electricity generation, known as agrivoltaics, has emerged as a promising strategy to diversify farm income, enhance energy security, and reduce land-use competition. Previous studies have demonstrated agrivoltaic benefits for specialty crops such as vegetables, berries, and pollinator habitats, where partial shading from solar panels can improve crop yields by reducing heat and water stress [1], [2]. However, comparatively little research has focused on the implications of agrivoltaic systems for **commodity crops** such as corn, which occupies about one-third of U.S. cropland [3].

Corn is a critical staple crop in the United States, accounting for the largest share of national grain production and forming the economic backbone of many rural communities [3]. Its integration with solar energy systems, however, poses particular challenges. Corn is generally regarded as shade-intolerant, with experimental studies suggesting a roughly linear reduction in yield with increased shading levels [4]. Yet these studies often derive from non-U.S. contexts and vary widely in their experimental setups, crop varieties, and climatic conditions, limiting their direct applicability to large-scale agrivoltaic deployment in the Midwest and other major U.S. corn-producing regions.

Moreover, shading from solar panels can also induce **potentially counterbalancing effects**. Partial shading may reduce soil evaporation and thus improve water retention, and mitigate heat stress during periods of extreme temperature, thereby offsetting some yield loss associated with reduced solar radiation [5]. Quantifying the net impact of these competing factors is critical for determining the economic viability and optimal design of agrivoltaic systems on commodity crop farmland.

In previous work, we developed a modeling framework integrating crop growth, soil water balance, and photovoltaic models to analyze the impacts of agrivoltaic shading on corn yields at the county level, using Renville County, MN, which often ranks among the top 20 corn-producing counties in the U.S., as a case study [6]. That study suggested that corn yields decreased by approximately 0.55% for every 1% increase in shading, with soil moisture benefits partially mitigating the effects of reduced radiation. However, the generalizability of this finding across diverse growing environments, soil types, and management practices in the U.S. remains unexplored.

The objectives of this study are threefold:

1. To **quantify the expected corn yield loss** under sparse solar arrays across the top 1000 corn-producing counties in the U.S.;
2. To **evaluate the counterbalancing effects** of yield loss due to shading versus increased soil moisture retention;
3. To **assess local variations in yield response and drought stress reduction** using county-level model calibration against 20 years of USDA-reported yield data.

By extending our modeling framework to a national scale, this study provides a systematic, county-level assessment of agrivoltaic impacts on corn production in the U.S. Our findings may inform regional agrivoltaic design, policy incentives, and economic feasibility analyses for dual-use deployment in commodity crop agriculture.

2. Modeling Framework

The modeling framework used in this study builds upon our previous work [6], which integrated a crop growth model [7], a soil water balance model [8], and a photovoltaic (PV) model to assess agrivoltaic impacts on corn yields at the county level. In this work, we extended the framework to analyze the top 1000 corn-producing counties in the United States, according to absolute corn production in 2022 accessed from the USDA Quick Stats Database [9], enabling a comprehensive national-scale assessment. Counties considered are shown in Figure 1. The vast majority of these counties are located in the “Corn Belt”, which includes Iowa, Illinois, Indiana, Ohio, Nebraska, Kansas, Minnesota, and Missouri.

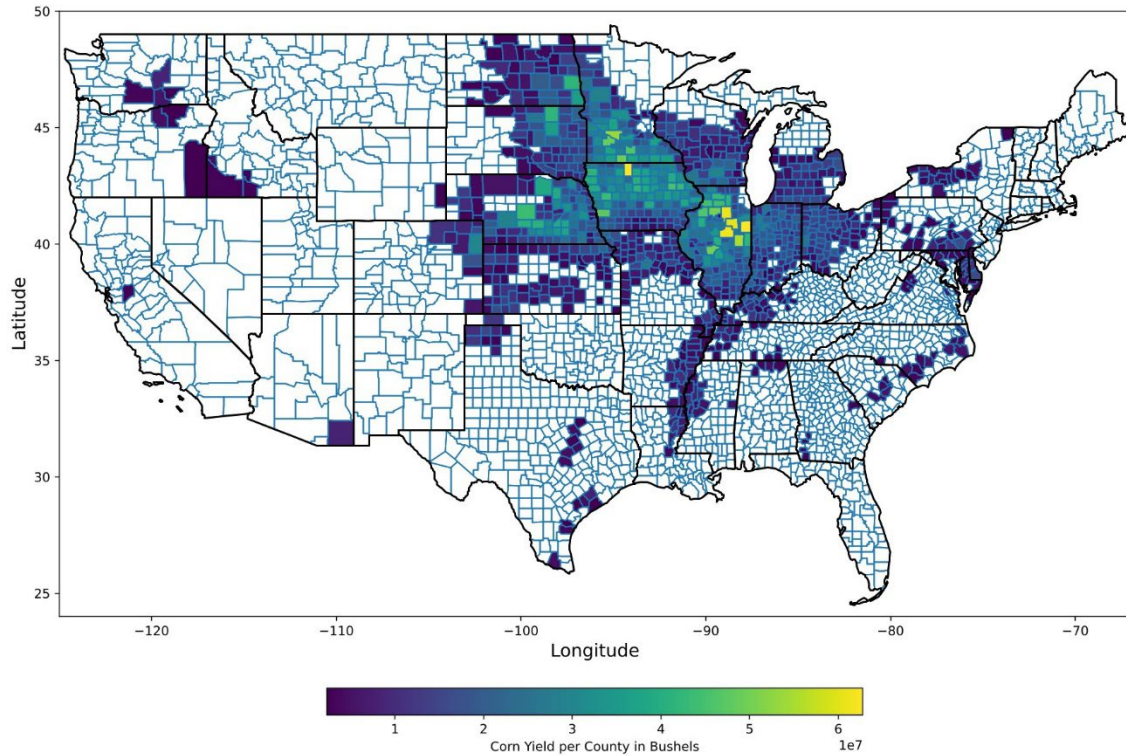


Figure 1. Top 1000 corn-producing counties in the United States.

For each county, we calibrated the crop growth model by comparing its predictions to 20 years (2003–2022) of USDA-reported yield data. This approach captures county-specific variations in soil characteristics, weather patterns, management practices, and genetic improvements. Model calibration was performed for each county by optimizing key crop parameters using the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm to minimize the difference between observed and predicted yields, resulting in a “county-average cultivar” representative of local planting practices.

The crop growth component is based on the SIMPLE model [7], which simulates daily biomass accumulation as a function of intercepted solar radiation, modified by stress factors including high temperature, low plant-available water, and low daily mean temperature. Soil water balance was modeled following the approach by Woli et al. [8], calculating plant-available water based on precipitation, evapotranspiration, runoff, and drainage. Drought stress was quantified using the Agricultural Reference Index for Drought (ARID), which ranges from 0 (saturated soil) to 1 (extreme drought). Generally, a reduction in drought stress will enhance yield. Other potentially important factors, such as the reduction of the leaf temperature due to shading, are not accounted for within the SIMPLE model.

Daily direct and diffuse solar irradiation data were obtained from the NASA POWER database [10] to calculate both crop-available radiation and PV electricity generation. In this study, we assumed static, south-facing PV arrays with low average shading levels ranging from 0% to 10%, because corn is a shade-intolerant crop and producers may be unwilling to tolerate yield losses associated with larger ground coverage ratios. As yield loss depends linearly on the amount of shading within the SIMPLE model, shading was assumed to be uniform over the crop and not spatially resolved under the solar array. Unlike our previous work that focused on a single county, this study enabled the evaluation of **county-specific drought stress responses, yield sensitivities to shading, and historical yield trends**, capturing the effects of improved corn seed varieties and management practices at regional scales.

All data processing and simulations were implemented in Python using Numpy, Scipy, and Pandas packages. Model calibration and simulation outputs were aggregated to generate national-scale yield loss estimates, drought stress mitigation trends, and insights into spatial variability in agrivoltaic viability across the United States.

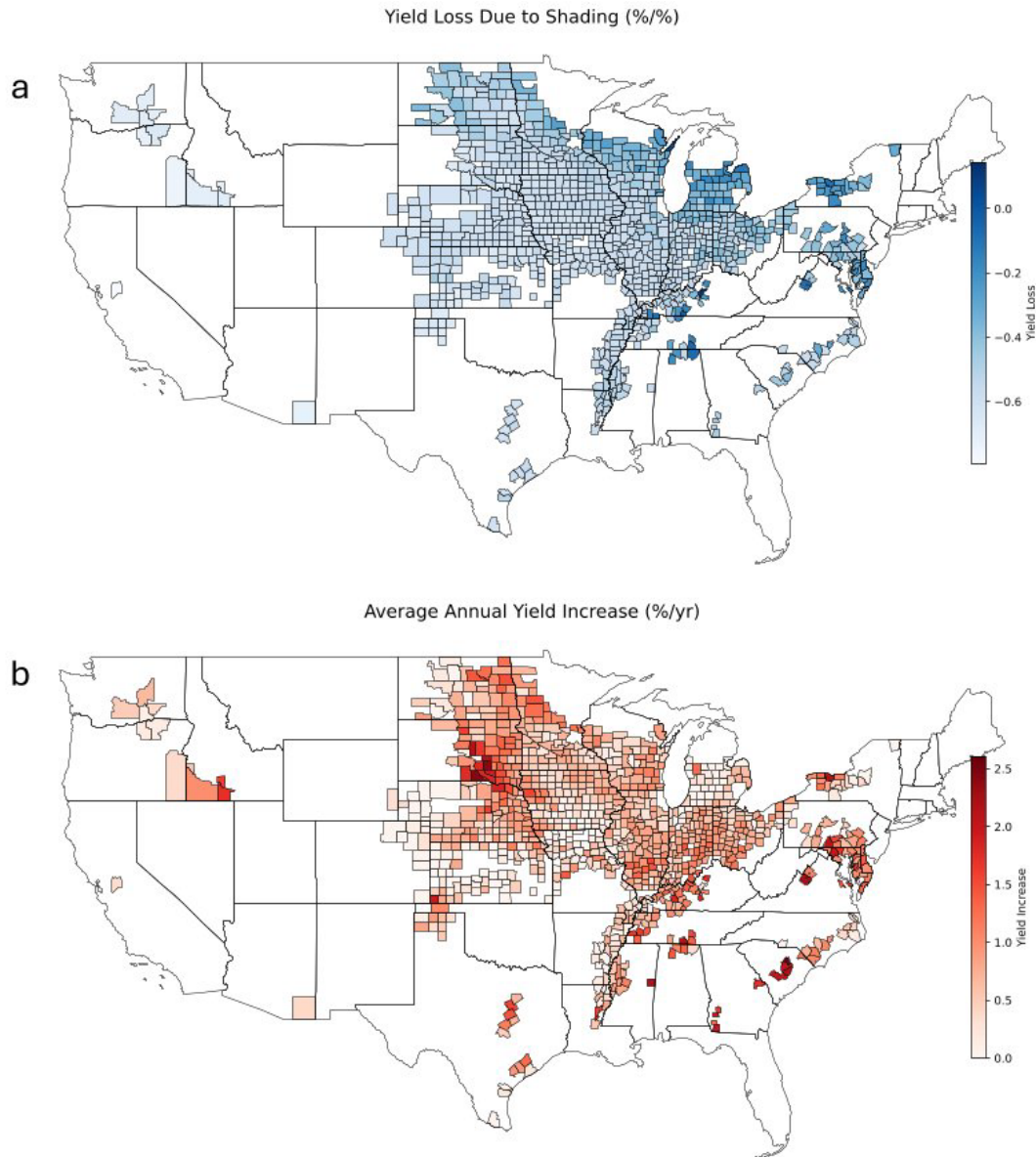


Figure 2. (a) Predicted corn yield loss for different counties considered in this study. (b) Average annual yield increase due to improved crop varieties and management practices.

3. Results

3.1 Predicted Yield Loss Due to Shading

Simulations were conducted for shading levels ranging from 0% to 10% to assess the impact of agrivoltaic arrays on county-level corn yields. The maximum shading level considered here was determined based on limited interviews with farmers about what they consider an acceptable level of shading from a farmer's perspective. Results showed that corn yield decreased nearly linearly with increasing shading, with an **average yield reduction of approximately 0.55% per 1% shading** (Figure 2a). This finding is consistent with our previous Renville

County study [6], indicating that the shading–yield relationship is broadly applicable across diverse U.S. growing environments. The degree of yield reduction varied moderately between counties due to differences in local climatic conditions, soil water-holding capacity, and baseline drought stress.

The model correctly captures an annual $\sim 0.5\%$ increase in historic corn yields, due to improved crop varieties and management practices, Figure 2b. However, local variations range from almost 0% annual increase at the low end to $\sim 2.5\%$ at the high end.

3.2 Impact of Shading on Drought Stress

In addition to reducing available radiation, shading from PV panels was found to mitigate drought stress, as quantified by the Agricultural Reference Index for Drought (ARID). Averaged over 20 growth seasons, the season-averaged ARID index varied from about 0.3 to 0.7, with some counties having season-averaged ARID indices as high as 0.9, Figure 3a. It should be noted that in order to describe a seasonal growth behavior, the seasonal ARID index is a more meaningful measure. Our analysis also does not include the impact of irrigation, because these data were not available.

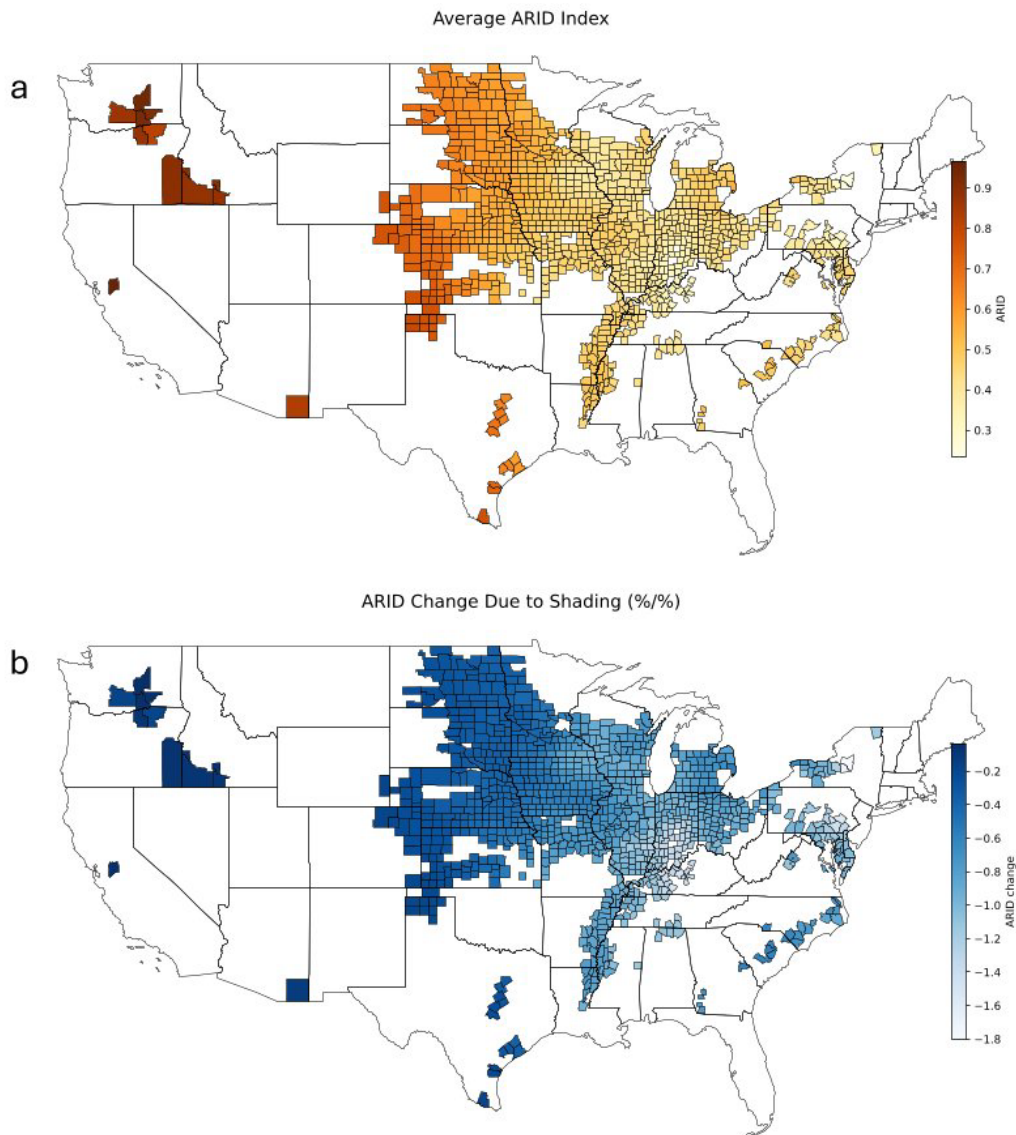


Figure 3. (a) Seasonally averaged ARID index for 20 growth periods from 2003-2022. (b) Fractional reduction in ARID index due to shading by PV panels.

Across all counties, the ARID index decreased by an average of **approximately 0.8% per 1% shading** (Figure 3b), reflecting improved soil moisture retention under partial shading. This reduction in drought stress was more pronounced in counties with lower baseline precipitation, where evapotranspiration losses are a major constraint to crop growth.

3.3 Drought Resistance of Modern Corn Varieties

An important outcome of the county-level calibration was the observation that modern corn varieties exhibited significantly higher drought tolerance compared to the McCurdy 84aa cultivar used in the original SIMPLE model [7]. The calibrated S_{water} parameter, representing sensitivity to soil water deficits, was notably lower for most counties, suggesting that recent advances in seed genetics have improved drought resilience. Consequently, while increased soil moisture under shading remains beneficial, its impact on yield loss mitigation is smaller in drought-resistant modern cultivars.

4. Conclusions

This study expanded our previous agrivoltaic modeling framework to the top 1000 corn-producing counties in the U.S., calibrating each county model with 20 years of USDA yield data to capture local conditions and management practices.

The results showed that corn yields decrease by approximately 0.55% per 1% shading, while drought stress (ARID index) decreases by ~0.8% per 1% shading due to improved soil moisture retention. The model also captured a ~0.5% per year historical yield increase, reflecting genetic and management improvements. Notably, modern corn varieties exhibited greater drought tolerance than legacy cultivars, reducing the extent to which soil moisture gains offset yield losses.

These findings highlight that while shading benefits soil moisture, net yield reductions remain notable, potentially impacting farm profitability in commodity crop systems. The framework developed here supports regional agrivoltaic planning by quantifying trade-offs between crop production and solar energy generation. Future work will include additional crops such as soybeans, and a larger set of solar configurations to further inform dual-use agricultural energy strategies.

Data availability statement

Data will be provided upon reasonable request to the corresponding author.

Underlying and related material

No supplementary materials are provided.

Author contributions

U.R.K: Conceptualization, Methodology, Investigation, Software, Writing – original draft, Writing – review & editing. V.E.F.: Conceptualization, Methodology, Writing – review & editing.

Competing interests

The authors declare no competing interests.

Funding

This work was supported by the University of Minnesota under the Ronald L. and Janet A. Christenson Chair in Renewable Energy. The authors acknowledge partial support from the Institute on the Environment 2024 Impact Goal grant IG-119.

References

- [1] M. I. Hermelink, B. Maestrini, and F. J. De Ruijter, "Berry shade tolerance for agrivoltaics systems: A meta-analysis," *Sci. Hortic.*, vol. 330, p. 113062, Apr. 2024, doi: [10.1016/j.scienta.2024.113062](https://doi.org/10.1016/j.scienta.2024.113062).
- [2] M. Trommsdorff *et al.*, "Combining food and energy production: Design of an agrivoltaic system applied in arable and vegetable farming in Germany," *Renew. Sustain. Energy Rev.*, vol. 140, p. 110694, Apr. 2021, doi: [10.1016/j.rser.2020.110694](https://doi.org/10.1016/j.rser.2020.110694).
- [3] J. K. Bond, "Corn and Other Feed Grains - Feed Grains Sector at a Glance," US Department of Agriculture - Economic Research Service. [Online]. Available: <https://www.ers.usda.gov/topics/crops/corn-and-other-feed-grains/feed-grains-sector-at-a-glance>
- [4] M. Laub, L. Pataczek, A. Feuerbacher, S. Zikeli, and P. Högy, "Contrasting yield responses at varying levels of shade suggest different suitability of crops for dual land-use systems: a meta-analysis," *Agron. Sustain. Dev.*, vol. 42, no. 3, p. 51, Jun. 2022, doi: [10.1007/s13593-022-00783-7](https://doi.org/10.1007/s13593-022-00783-7).
- [5] D. Soto-Gómez, "Integration of Crops, Livestock, and Solar Panels: A Review of Agrivoltaic Systems," *Agronomy*, vol. 14, no. 8, p. 1824, Aug. 2024, doi: [10.3390/agronomy14081824](https://doi.org/10.3390/agronomy14081824).
- [6] U. Kortshagen and V. Ferry, "Modeling the Agrivoltaic Potential for Land-Intensive Commodity Crops," *AgriVoltaics Conf. Proc.*, vol. 3, Apr. 2025, doi: [10.52825/agripv.v3i.1368](https://doi.org/10.52825/agripv.v3i.1368).
- [7] C. Zhao *et al.*, "A SIMPLE crop model," *Eur. J. Agron.*, vol. 104, pp. 97–106, Mar. 2019, doi: [10.1016/j.eja.2019.01.009](https://doi.org/10.1016/j.eja.2019.01.009).
- [8] P. Woli, J. W. Jones, K. T. Ingram, and C. W. Fraisse, "Agricultural Reference Index for Drought (ARID)," *Agron. J.*, vol. 104, no. 2, pp. 287–300, Mar. 2012, doi: [10.2134/agronj2011.0286](https://doi.org/10.2134/agronj2011.0286).
- [9] "USDA National Agricultural Statistics Service." [Online]. Available: <https://quickstats.nass.usda.gov>
- [10] "NASA Prediction Of Worldwide Energy Resources (POWER)." [Online]. Available: <https://power.larc.nasa.gov/api/pages/>